A Study of Fractal Image Compression with Coefficient Quantization

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Abstract

In recent years fractal image compression (FIC) have gained more interest because of their capability to achieve high compression ratios while maintaining very good quality of the reconstructed image. Many encoding method were proposed to reduce the encoding time, but coefficient quantization in FIC was proposed little. In this paper, we discuss the coefficient quantization in FIC and the parameter how to affect quality of reconstruct image.

1 Introduction

The fractal transform for image compression was introduced by Barnsley and Demko [1]. The idea of the fractal image compression (FIC) is based on the assumption that the image redundancies can be efficiently exploited by mean s of block self-affine transformations [2,3]. The image is represented through a piecewise linear contractive function F and is reconstructed by iteratively applying F to a randomly chosen starting image. This technique is called the Iterated Function System (IFS). FIC has been used in many applications of image processing such as feature extractions, image watermarking, image retrievals and image denoising [4-8]. Fractal image coding has advantages like fast decoding, resolution independent decoding, high compression ratio and highly reconstructed image quality. The long encoding time has been a major drawback of fractal image coding. Therefore, plenty of research focused on how to improve the speed of fractal image coding, and almost all of them explored how to reduce the amount of domain blocks in domain pool [9-12]. Different from reducing the domain blocks in domain pool, some other researchers focused on making the search faster [13,14]. Fractal image encoding have advantages like fast decoding, resolution independent decoding. Through the long encoding time has been a major drawback to prevent it from practical usage. In order to obtain very high compression ratio (CR), fractal code requires that coefficient representing the contractive function will be represented by very few bit: this fact causes a degradation of the image which can be strongly reduced through a careful choice of quantization functions. In recent years, many encoding strategies were proposed to reduce the encoding time [15-21], but coefficient quantization in FIC was proposed little [22].

2 Fractal Image Compression

The IFS consists of a set contraction mappings w_i. A transformation $w_i : X \to X$ for $1 \le i \le K$, defines in a complete metric space (X, h), satisfies:

$$h(w_i(x), w_i(y)) \le s_i h(x, y) \quad \forall x, y \in X$$

where X and h denote the space and its corresponding metric, respectively, and K denotes the total number of contraction mappings. The mapping w_i is referred to as contractive with contractively s_i, where $0 \le s_i < 1$. The fractal transformation associated with an IFS is defined as:

$$W(B)\bigcup_{i=1}^{K} w_i(B) \quad \forall B \in H(X)$$

where B is any element of the space of non-empty compact subsets of X. If w_i is contractive for every i, then W is contractive and generates a unique attractor A such that:

$$W(A) = A$$

$$\lim_{r\to\infty}(B) = A \quad \forall B \in H(X)$$

with contractively $s = \max\{s_i : 1 \le i \le K\}$, where A is called the attractor of IFSZ and the transformations are usually chosen to be affine transformations.

Additionally, the Collage Theorem [3]:

$$h(B, A) \le (1-s)^{-1} h(B, W(B)) \quad \forall B \in H(X)$$

provides justification for independent construction of each mapping, where h is the Hausdorff Distance [3]. That is to say that we can guarantee that A and B will be sufficiently close if we can make B and $W(B)$ close enough. Then we can reconstruct an image by generating the attractor of the IFS without direct information on an input image. The principle of fractal coding is based on the IFS developed by Barnsley [3]. Jacquin was the first to propose a block-based fractal coding scheme for gray-level images [23].

For Fig.1, it is almost impossible for finding the fractal code of IFS. However, we can find some similarities between blocks of sub-images. In Fig. 1, there are two pairs of blocks, which are similar to each other. Taking into account the gray level as a third dimension, the transformation function will become Equation (1):

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = w_i \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & p \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ q \end{bmatrix} \quad (1)$$

where z and z' are the gray levels, a, b, c, and d are coordinates of this transformation, (t_x, t_y) is the offset of the transformation, p controls the contrast and q controls the brightness of the transformation.

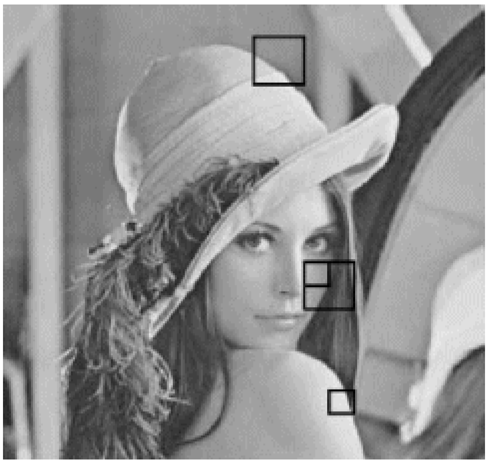

Fig. 1 There are some similarities between the sub-images in the image Lena..

The basic fractal coding scheme is to divide an image (with size $N \times N$) into non-overlapped block called range blocks $R = \{R_1, R_2, ..., R_u\}$ and overlapped block called domain blocks $D = \{D_1, D_2, ..., D_v\}$, where u is the number of R and v is the number of D. The size of domain block is twice or more times of that of range block. The transform τ consisting of geometric part and massic part [24] using partitioned iterated function system (PIFS). The geometric part denotes the down sampling operation for the domain block D. The massic part represents eight symmetries on transformed domain block such as (1) identity (2) rotation through +90 degree (3) rotation through +180 degree (4) rotation through +270 degree (5) reflection about mid-vertical axis (6) reflection about mid-horizontal axis (7) reflection about first diagonal (8) reflection about second diagonal. The transform τ is defined as

$$\tau = p \times D + q \times I$$

where s is the scaling factor, o is the luminance offset factor and I is the identity block having the same size as D. Let d_i and r_i are elements of domain block D and range block R, respectively. The distance measure d between the range block and the domain block is often calculated by the mean square error (MSE)

$$d(R, D) = \sum_{i=1}^{n} [(pd_i + q) - r_i]^2 \quad (2)$$

The least-squares results of optimal p and q are given by

$$p_k = \frac{N^2 < u_k, v > - \sum_{i=o}^{N-1} \sum_{j=0}^{N-1} u_k(i,j) \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} v(i,j)}{N^2 < u_k, u_k > -(\sum_{i=0}^{N-1} \sum_{j=0}^{N-1} u_k(i,j))^2} \quad (3)$$

$$q_k = \frac{1}{N^2} \left[\sum_{i=0}^{N-1} \sum_{j=0}^{N-1} v(i,j) - p_k \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} u_k(i,j) \right] \quad (4)$$

Firstly, divide the image into range block and domain block. Secondly, the parameters p and q, and the mapping location of domain d are stored when the best matched domain block has been found. The condition is that the current MSE between range block and domain block is the minimum one.

3 Coefficient Quantization

The compression ratio of IFS regards the number of bit of coefficient. When the coefficient represented by very few bit, the IFS code obtain very high compression ratio. The coefficient represented not only affects the compression ratio, but also affects the quality of decoding image. The fractal code include i, j, k, p, and q. The i and j coefficients represent the position of domain block, and the bits of these coefficients be decided by image size and domain block size. The k coefficient represent one of the eight isometric transformations (identity, reflection about x and y axes, about first and second diagonals, $+90^\circ$, $+180^\circ$, $+270^\circ$ rotations around the center) to domain block, and it used 3 bits to represent. p controls the contrast and q controls the brightness of domain block, and the bit of these coefficient can to affect compression ratio and decoding image quality.

In order to map the continuous values $p \in R = [L_p, U_p]$ and $q \in R = [L_q, U_q]$ into the quantized value $\tilde{p}$ and $\tilde{q}$ which represented with b_p and b_q bits, it used linear quantization function to transform coefficient. Each range can be partitioned into k intervals $I_h = (L_h, U_h]$, resulting $L_h = U_{h-1} (h = 2,3,...,k)$. We choose to associate it to the quantized value

$$\tilde{p} = \frac{U_h + L_h}{2} \quad (5)$$

when given any number p belonging to an interval I_h. The q coefficient has same process to quantize to $\tilde{q}$.

There are two methods for coefficient quantization. First method, it quantize p and q coefficients when finish search all i, j, k, p, and q for each range block. Second method, it quantize p

and q coefficients and use $\tilde{p}$ and $\tilde{q}$ to evaluate the root mean square (RMS) error for all the possible pairs of range blocks and domain blocks.

4 Experiments

In this session, we show the parameter how to affect quality of reconstruct image. The PSNR given by $PSNR = 10 \cdot \log_{10}(255^2 / MSE)$ is used to measure the quality of the retrieved images. The tested pattern is the gray level Lena images of size 256×256. In Table 1 we report, for different range of p and q coefficient when p used 5 bits and q used 7 bits. The PSNR is similar to different range of p and q in method 2. In Table 2 we report, for different range of p and q coefficient when p used 3 bits and q used 3 bits. The PSNR is better then others when the range of p and q coefficient is small. In other words, the range of p and q coefficient needs small, p and q coefficient used few bits especially. The method 1 has the same direction even if the PSNR of method 1 is bad then the PSNR of method 2.

5 Conclusion

In this paper, we discuss the coefficient quantization in fractal image compression. Fractal code requires that coefficient representing the contractive function will be represented by very few bit when want to raise compression ratio. The range of p and q coefficients will affect decoding image quality when the coefficients use few bits. Experimental results show that the PSNR for the retrieved image when the range of p and q coefficient is small.

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(a) $p=[-2,2]$, $q=[-256,256]$, PSNR=28.54

(b) $p=[-2,2]$, $q=[-512,512]$, PSNR=28.56

(c) $p=[-4,4]$, $q=[-256,256]$, PSNR=28.36

(d) $p=[-4,4]$, $q=[-512,512]$, PSNR=28.38

Fig. 2 Coefficient quantization by method 2, p used 5 bits and q used 7 bits.

(a) $p = [-2,2]$, $q = [-256,256]$, PSNR=27.14

(b) $p = [-2,2]$, $q = [-512,512]$, PSNR=26.71

(c) $p = [-4,4]$, $q = [-256,256]$, PSNR=26.64

(d) $p = [-4,4]$, $q = [-512,512]$, PSNR=24.83

Fig. 3 Coefficient quantization by method 2, p used 3 bits and q used 3 bits.

Table 1

p range		*q* range		Method 1	Method 2
L	*U*	*L*	*U*	PSNR	PSNR
-2	2	-256	256	20.67	28.54
-2	2	-512	512	18.61	28.56
-4	4	-256	256	19.48	28.36
-4	4	-512	512	22.19	28.38
-8.59 (min)	9.031 (max)	-1776 (min)	1212.36 (max)	18.11	27.82

Table 2

p range		*q* range		Method 1	Method 2
L	*U*	*L*	*U*	PSNR	PSNR
-2	2	-256	256	16.49	27.14
-2	2	-512	512	13.36	26.71
-4	4	-256	256	14.05	26.64
-4	4	-512	512	12.72	24.83
-8.59 (min)	9.031 (max)	-1776 (min)	1212.36 (max)	9.08	13.90